# Trigonometry

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**ANSWER KEY — COMPLETE WORKED SOLUTIONS**

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 5x + 4$ in writing rules for relationships, evaluate $f(5)$.

Final Answer: $29$. Substitute $x = 5$: $f(5) = 5(5) + 4 = 29$.

**2** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 6x + 5$ in substitution, evaluate $f(2)$.

Final Answer: $17$. Substitute $x = 2$: $f(2) = 6(2) + 5 = 17$.

**3** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for substitution consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: $24.5$. The sum is $18 + 22 + 26 + 32 = 98$, and dividing by $4$ gives $24.5$.

**4** [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 2x + 6$ in collecting like terms, evaluate $f(2)$.

Final Answer: $10$. Substitute $x = 2$: $f(2) = 2(2) + 6 = 10$.

**5** [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for collecting like terms consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: $16.5$. The sum is $10 + 14 + 18 + 24 = 66$, and dividing by $4$ gives $16.5$.

**6** [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the model $f(x) = 3x + 7$ in solving contextual problems with algebra, evaluate $f(2)$.

Final Answer: $13$. Substitute $x = 2$: $f(2) = 3(2) + 7 = 13$.

**7** [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for solving contextual problems with algebra consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: $17.5$. The sum is $11 + 15 + 19 + 25 = 70$, and dividing by $4$ gives $17.5$.

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## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

An equilateral triangle has side lengths of $12$ cm. An altitude is drawn from one vertex to the opposite side. Find the exact height and area of the triangle.

Solution: The altitude bisects the vertex angle into two $30^\circ$ angles and bisects the base into two equal segments of length $6$ cm, forming two congruent $30$–$60$–$90$ right triangles. The shorter leg opposite $30^\circ$ is $x = 6$. The altitude (height) is the longer leg opposite $60^\circ$: $h = x\sqrt{3} = 6\sqrt{3}$ cm. Now compute the area using Area $= \frac{1}{2}bh$: Area $= \frac{1}{2}(12)(6\sqrt{3}) = 36\sqrt{3}$ cm$^2$.

**9** [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

In a $45^\circ$–$45^\circ$–$90^\circ$ right triangle, the hypotenuse has length $10\sqrt{2}$ cm. What is the length of each leg, and what is the area of the triangle?

Step 1: Find leg length $s$: In a $45^\circ$–$45^\circ$–$90^\circ$ special right triangle, the hypotenuse is $s\sqrt{2}$: $s\sqrt{2} = 10\sqrt{2} \Rightarrow s = 10$ cm. Step 2: Calculate the area: Since both legs form the base and height: Area $= \frac{1}{2}s^2 = \frac{1}{2}(10^2) = \frac{1}{2}(100) = 50$ cm$^2$.

**10** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

[Right Triangles & Trigonometry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input $t$ when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

**11** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

[Trigonometry & Circular Measure · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input $t$ when $g(t) = 18$.

Substitute $t = 4$: $g(4) = 3(4) + 3 = \mathbf{15}$. Setting $3t + 3 = 18$ gives $3t = 15 \Rightarrow t = \mathbf{5}$.

**12** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

In isosceles triangle $\Delta PQR$, $PQ = PR$. The vertex angle $\angle P$ measures $40^\circ$. An angle bisector is drawn from base angle $\angle Q$ to point $D$ on side $PR$. What is the measure of $\angle QDR$?

Step 1: Find the base angles of $\Delta PQR$: Since $PQ = PR$, the base angles $\angle Q$ and $\angle R$ are equal: $m\angle Q = m\angle R = \frac{180^\circ - 40^\circ}{2} = \frac{140^\circ}{2} = 70^\circ$. Step 2: Find the bisected angle measure: $QD$ bisects $\angle PQR$, so: $m\angle DQR = \frac{70^\circ}{2} = 35^\circ$. Step 3: Apply the triangle angle sum to $\Delta QDR$.

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**13** [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Solve the equation $\sqrt{2x+7} = x+2$ for all real values of $x$.

Square both sides of the equation: $(\sqrt{2x+7})^2 = (x+2)^2 \Rightarrow 2x+7 = x^2+4x+4$. Rearrange into standard quadratic form: $x^2+2x-3=0 \Rightarrow (x+3)(x-1)=0 \Rightarrow x=1$ or $x=-3$. Now check both candidate roots in the original equation:For $x=1$: LHS = $\sqrt{2(1)+7} = \sqrt{9} = 3$; RHS = $1+2=3$. (Valid ✓)For $x=-3$: LHS = $\sqrt{2(-3)+7} = \sqrt{1} = 1$; RHS = $-3+2=-1$. Since $1eq-1$, $x=-3$ is extraneous.The only real solution is.

**14** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A circle is inscribed inside a square of side length $10$ cm. A point is chosen at random inside the square. What is the exact probability that the point lands outside the circle (in the four shaded corners)?

Solution: The diameter of the inscribed circle equals the side of the square ($10$ cm), so the radius is $r = 5$ cm. Step 1 (Total area): $\text{Area(square)} = 10^2 = 100$ cm$^2$. Step 2 (Circle area): $\text{Area(circle)} = \pi(5^2) = 25\pi$ cm$^2$. Step 3 (Corner area): $\text{Area(outside)} = 100 - 25\pi$. Step 4 (Probability): $P(\text{outside}) = \frac{100-25\pi}{100} = 1 - \frac{\pi}{4} \approx 1 - 0.7854 = 0.2146$ (21.5%).

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Algebra & Graphs

The function $f$ is defined by $f(x) = x^2 - 6x + 13$ for $x \geq 3$. (a) State the range of $f$. (b) Find an expression for $f^{-1}(x)$ and state the domain and range of $f^{-1}$.

(a) Complete the square: $f(x) = (x-3)^2 - 9 + 13 = (x-3)^2 + 4$. Since $x \geq 3$, $(x-3)^2 \geq 0$, so the minimum value is $f(3) = 4$. The range of $f$ is $f(x) \geq 4$. (b) Set $y = (x-3)^2 + 4$. Rearrange for $x$: $y - 4 = (x-3)^2 \Rightarrow x - 3 = +\sqrt{y-4}$ (since $x \geq 3$) $x = 3 + \sqrt{y-4}$ Therefore, $f^{-1}(x) = 3 + \sqrt{x-4}$. Domain of $f^{-1}$ is $x \geq 4$ (the range of $f$), and range of $f^{-1}$ is $f^{-1}(x) \geq 3$ (the domain of $f$).

**16** [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Coordinate Geometry

Functions $g$ and $h$ are defined by $g(x) = 2x - 1$ for $x \in \mathbb{R}$, and $h(x) = \frac{6}{x-2}$ for $xeq2$. (a) Find an expression for $hg(x)$ in simplest form. (b) Solve the equation $hg(x) = g(x)$.

(a) Substitute $g(x)$ into $h$: $hg(x) = h(2x-1) = \frac{6}{(2x-1)-2} = \frac{6}{2x-3}$ $\left(xeq\frac{3}{2}\right)$ (b) Set $hg(x) = g(x)$: $\frac{6}{2x-3} = 2x-1 \Rightarrow 6 = (2x-1)(2x-3)$ Expand the right-hand side: $6 = 4x^2 - 8x + 3 \Rightarrow 4x^2 - 8x - 3 = 0$ Factor: $(2x-3)(2x+1) = 0$. Since $xeq\frac{3}{2}$ for $hg(x)$ to be defined, the value $x = \frac{3}{2}$ is excluded. Therefore, the unique solution is $2x + 1 = 0 \Rightarrow x = -\frac{1}{2}$.

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**17** **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)** • **Cross-Lesson: 4.01 Mensuration**

Find the exact area of the region enclosed between the curve $y = 4x - x^2$ and the line $y = x$.

Find points of intersection: $4x - x^2 = x \Rightarrow 3x - x^2 = 0 \Rightarrow x(3 - x) = 0$ The curves intersect at $x = 0$ and $x = 3$. Between $0$ and $3$, the curve $y = 4x - x^2$ lies above the line $y = x$. The enclosed area is: $A = \int_0^3 [(4x - x^2) - x]\,dx = \int_0^3 (3x - x^2)\,dx$ Integrate: $A = \left[\frac{3x^2}{2} - \frac{x^3}{3}\right]_0^3 = \left(\frac{3(9)}{2} - \frac{27}{3}\right) - 0 = \frac{27}{2} - 9 = \frac{27-18}{2} = \frac{9}{2} = 4.5$

**18** **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 5.01 Geometry**

Evaluate the numerical value of $16^{-3/4} + \left(\frac{27}{8}\right)^{-2/3}$.

1. Evaluate the first term: $16^{-3/4} = \frac{1}{16^{3/4}} = \frac{1}{(\sqrt[4]{16})^3} = \frac{1}{2^3} = \frac{1}{8}$. 2. Evaluate the second term: inverting the fraction removes the negative exponent: $\left(\frac{27}{8}\right)^{-2/3} = \left(\frac{8}{27}\right)^{2/3} = \frac{(\sqrt[3]{8})^2}{(\sqrt[3]{27})^2} = \frac{2^2}{3^2} = \frac{4}{9}$. 3. Combine with a common denominator of $72$: $\frac{1}{8} + \frac{4}{9} = \frac{9}{72} + \frac{32}{72} = \frac{41}{72}$.

**19** **[Challenging]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)** • **Cross-Lesson: 7.01 Transformations & Vectors**

Two similar cylinders have surface areas of $64\pi$ cm$^2$ and $144\pi$ cm$^2$. If the volume of the smaller cylinder is $96\pi$ cm$^3$, find the volume of the larger cylinder.

Step 1: Find the area ratio and linear scale factor $k$: $\frac{\text{Area}_2}{\text{Area}_1} = \frac{144\pi}{64\pi} = \frac{9}{4} = k^2 \Rightarrow k = \sqrt{\frac{9}{4}} = \frac{3}{2}$ Step 2: Compute the volume ratio $k^3$: $\frac{\text{Volume}_2}{\text{Volume}_1} = k^3 = \left(\frac{3}{2}\right)^3 = \frac{27}{8}$ Step 3: Calculate the volume of the larger cylinder: $\text{Volume}_2 = 96\pi \times \frac{27}{8} = 12\pi \times 27 = 324\pi$ cm$^3$

**20** **[Challenging]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)** • **Cross-Lesson: 8.01 Probability & Statistics**

A curve has equation $y = 2x^3 - 9x^2 + 12x + 1$. (a) Find the coordinates of both stationary points. (b) Determine the nature of each stationary point using the second derivative $\frac{d^2y}{dx^2}$.

(a) Find $\frac{dy}{dx}$: $\frac{dy}{dx} = 6x^2 - 18x + 12$ Set $\frac{dy}{dx} = 0$: $6(x^2 - 3x + 2) = 0 \Rightarrow 6(x - 1)(x - 2) = 0$, giving $x = 1$ and $x = 2$. When $x = 1$: $y = 2(1) - 9(1) + 12(1) + 1 = 6$. Point is $(1, 6)$. When $x = 2$: $y = 2(8) - 9(4) + 12(2) + 1 = 16 - 36 + 24 + 1 = 5$. Point is $(2, 5)$. (b) Second derivative: $\frac{d^2y}{dx^2} = 12x - 18$ At $x = 1$: $\frac{d^2y}{dx^2} = 12(1) - 18 = -60$, so $(2, 5)$ is a local minimum.

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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**21** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 9$ in introduction to variables, evaluate $f(2)$.

**a.** 15 ✓

**b.** 18

**c.** 6

**d.** 19

Substitute $x = 2$: $f(2) = 3(2) + 9 = 15$.

**22** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 4x + 3$ in building algebraic expressions, evaluate $f(2)$.

**a.** 11 ✓

**b.** 15

**c.** 8

**d.** 15

Substitute $x = 2$: $f(2) = 4(2) + 3 = 11$.

**23** [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for introduction to variables consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

**a.** 19.5

**b.** 20.5

**c.** 21.5 ✓

**d.** 23.5

The sum is $15 + 19 + 23 + 29 = 86$, and dividing by $4$ gives $21.5$.

**24** [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for building algebraic expressions consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

**a.** 20.5

**b.** 21.5

**c.** 22.5 ✓

**d.** 24.5

The sum is $16 + 20 + 24 + 30 = 90$, and dividing by $4$ gives $22.5$.

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**25** [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In triangle $ABC$, angle $A = 30^\circ$, angle $B = 90^\circ$, and side $a = 8$ cm (opposite $A$). Find side $b$ (opposite $B$), to 2 dp.

**a.** 18.0 cm

**b.** **16.0 cm ✓**

**c.** 17.0 cm

**d.** 15.0 cm

$\frac{a}{\sin A} = \frac{b}{\sin B} \Rightarrow b = \frac{8\sin(90^\circ)}{\sin(30^\circ)} = 16.0$ cm.

**26** [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

In triangle ABC: AB=7, angle A=35°, angle B=65°. Which rule to find BC?

**a.** Cosine rule

**b.** **Sine rule ✓**

**c.** Pythagoras

**d.** Area formula

Two angles and a non-included side → Sine rule.

**27** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In triangle $ABC$, $AB = 7$ cm, $AC = 9$ cm, and angle $A = 60^\circ$. Use the cosine rule to calculate side length $BC$.

**a.** **8.19 cm ✓**

**b.** 8.00 cm

**c.** 7.50 cm

**d.** 8.54 cm

Cosine rule: $a^2 = b^2 + c^2 - 2bc\cos A \Rightarrow BC^2 = 7^2 + 9^2 - 2(7)(9)\cos(60^\circ) = 49 + 81 - 126(0.5) = 130 - 63 = 67$. Then $BC = \sqrt{67} \approx 8.19$ cm. [M1 for cosine rule setup, M1 for 67, A1 for 8.19]

**28** [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In a cuboid with a square base of 8 cm by 8 cm and height 10 cm, calculate the angle that the space diagonal makes with the base (to 1 decimal place).

**a.** **41.4° ✓**

**b.** 51.3°

**c.** 36.9°

**d.** 45.0°

Base diagonal = $\sqrt{8^2 + 8^2} = \sqrt{128} \approx 11.31$ cm. $\tan\theta = \frac{10}{\sqrt{128}} \approx 0.8839 \Rightarrow \theta \approx 41.4^\circ$. [M1 A1]

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## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

In right-angled triangle $ABC$, angle $C = 90^\circ$, $AB = 10$ cm and angle $A = 30^\circ$. Find the length of opposite side $BC$.

**a. 5 cm ✓**

b. 8.66 cm

c. 5.77 cm

d. 7.07 cm

Use $\sin A = \frac{\text{opposite}}{\text{hypotenuse}} \Rightarrow \sin(30^\circ) = \frac{BC}{10} \Rightarrow BC = 10 \times 0.5 = 5$ cm. [M1 for sin(30)=BC/10, A1 for 5]

**30** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the area of triangle $ABC$ where $a = 6$ cm, $b = 8$ cm, and included angle $C = 30^\circ$.

**a. 12 cm² ✓**

b. 24 cm²

c. 20.8 cm²

d. 48 cm²

Use the triangle area formula: $\text{Area} = \frac{1}{2}ab\sin C = \frac{1}{2}(6)(8)\sin(30^\circ) = 24 \times 0.5 = 12$ cm$^2$. [M1 for 0.5*a*b*sin(C), A1 for 12]

**31** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In triangle ABC, side b = 10 cm, angle B = 45°, and angle C = 60°. Calculate the length of side c to 1 decimal place.

**a. 12.2 cm ✓**

b. 8.2 cm

c. 14.1 cm

d. 10.5 cm

By the sine rule: $\frac{c}{\sin 60^\circ} = \frac{10}{\sin 45^\circ} \Rightarrow c = \frac{10\sin 60^\circ}{\sin 45^\circ} \approx 12.2$ cm. [M1 A1]

**32** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

In triangle PQR, side p = 5 cm, side q = 7 cm, and side r = 8 cm. Calculate angle R (to 1 decimal place).

**a. 81.8° ✓**

b. 60.0°

c. 75.5°

d. 98.2°

Cosine rule: $\cos R = \frac{5^2 + 7^2 - 8^2}{2(5)(7)} = \frac{25 + 49 - 64}{70} = \frac{10}{70} = \frac{1}{7} \Rightarrow R \approx 81.8^\circ$. [M1 A1]

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**33** [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A ship sails 15 km on a bearing of 060° and then 20 km on a bearing of 150°. How far is the ship from its starting point?

**a. 25 km ✓**

**b.** 35 km

**c.** 22.5 km

**d.** 30 km

The turn creates a right angle: $180^\circ - (150^\circ - 60^\circ) = 90^\circ$. Distance = $\sqrt{15^2 + 20^2} = \sqrt{625} = 25$ km. [M1 A1]

**34** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In triangle ABC, side a = 6 cm, side b = 10 cm, and side c = 14 cm. Find angle C.

**a. 120° ✓**

**b.** 60°

**c.** 150°

**d.** 135°

Cosine rule: $\cos C = \frac{6^2 + 10^2 - 14^2}{2(6)(10)} = \frac{36 + 100 - 196}{120} = \frac{-60}{120} = -0.5 \Rightarrow C = 120^\circ$. [M1 A1]

**35** [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

In triangle $ABC$, side $a = 8$ cm, angle $A = 30^\circ$, and angle $B = 45^\circ$. Find side $b$ to 1 decimal place.

**a. 11.3 cm ✓**

**b.** 13.8 cm

**c.** 9.5 cm

**d.** 12.0 cm

By the sine rule: $\frac{b}{\sin B} = \frac{a}{\sin A} \Rightarrow b = \frac{8 \sin 45^\circ}{\sin 30^\circ} = 11.3$ cm. [M1 A1]

**36** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In triangle $ABC$, side $b = 5$ cm, side $c = 8$ cm, and angle $A = 60^\circ$. Find side $a$ to 1 decimal place.

**a. 7.0 cm ✓**

**b.** 8.5 cm

**c.** 6.5 cm

**d.** 9.4 cm

Cosine rule: $a^2 = b^2 + c^2 - 2bc\cos A = 5^2 + 8^2 - 2(5)(8)\cos 60^\circ = 49 \Rightarrow a = \sqrt{49} = 7.0$ cm. [M1 A1]

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**37** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The bearing of ship $B$ from port $A$ is $065^\circ$. What is the back bearing of port $A$ from ship $B$?

**a.** $115^\circ$

**b.** $205^\circ$

**c.** $245^\circ$ ✓

**d.** $295^\circ$

To find the reverse bearing when the forward bearing is less than $180^\circ$, add $180^\circ$: $065^\circ + 180^\circ = 245^\circ$. [B1]

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## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Geometry

In right-angled triangle ABC, angle B = 90°, AB = 8 cm, and BC = 15 cm. Find the value of $\sin A$.

**a.** $\frac{15}{17}$ ✓

**b.** $\frac{8}{17}$

**c.** $\frac{15}{8}$

**d.** $\frac{8}{15}$

Hypotenuse $AC = \sqrt{8^2 + 15^2} = 17$ cm. $\sin A = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{15}{17}$. [M1 A1]

**39** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Transformations & Vectors

In right-angled triangle PQR, angle Q = 90°, PQ = 5 cm, and QR = 12 cm. Find the value of $\cos P$.

**a.** $\frac{5}{13}$ ✓

**b.** $\frac{12}{13}$

**c.** $\frac{5}{12}$

**d.** $\frac{12}{5}$

Hypotenuse $PR = \sqrt{5^2 + 12^2} = 13$ cm. $\cos P = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{5}{13}$. [M1 A1]

**40** [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 8.01 Probability & Statistics

Calculate the area of triangle $ABC$ where side $a = 6$ cm, side $b = 8$ cm, and included angle $C = 30^\circ$.

**a.** **12 cm²** ✓

**b.** 24 cm²

**c.** 19 cm²

**d.** 9 cm²

Formula: Area $= \frac{1}{2}ab\sin C = \frac{1}{2}(6)(8)\sin 30^\circ = \frac{1}{2}(48)(0.5) = 12$ cm². [M1 A1]

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**41** [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 9.01 Functions

In triangle ABC, side a = 7 cm, side b = 9 cm, and angle C = 40°. Calculate the area of triangle ABC to 1 decimal place.

**a. 20.2 cm² ✓**

b. 40.5 cm²

c. 31.5 cm²

d. 24.1 cm²

Area = $\frac{1}{2}ab\sin C = \frac{1}{2}(7)(9)\sin 40^\circ \approx 31.5 \times 0.6428 \approx 20.2$ cm². [M1 A1]

**42** [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Number

The bearing of point *B* from point *A* is $045^\circ$. What is the back-bearing of *A* from *B*?

**a. 225° ✓**

b. 270°

c. 315°

d. 015°

Back-bearing differs by $180^\circ$: since $45^\circ < 180^\circ$, back-bearing = $45^\circ + 180^\circ = 225°$. [B1]

**43** [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 10.01 Sets & Venn Diagrams

In right-angled triangle XYZ, angle Y = 90°, angle X = 30°, and hypotenuse XZ = 18 cm. What is the length of side YZ?

**a. 9 cm ✓**

b. 18 cm

c. $9\sqrt{3}$ cm

d. 4.5 cm

Side opposite 30° is half the hypotenuse: $YZ = 18 \times \sin 30^\circ = 18 \times 0.5 = 9$ cm. [M1 A1]

**44** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Algebra & Graphs

From the top of a vertical cliff 40 m high, the angle of depression of a boat out at sea is 25°. How far is the boat from the base of the cliff (to 1 decimal place)?

**a. 85.8 m ✓**

b. 95.2 m

c. 18.7 m

d. 44.1 m

$\tan 25^\circ = \frac{40}{d} \Rightarrow d = \frac{40}{\tan 25^\circ} \approx 85.8$ m. [M1 A1]

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**45** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Coordinate Geometry

Given that $\sin x = 0.6$ where $90^\circ < x < 180^\circ$ (obtuse angle), find the value of $\cos x$.

**a.** **-0.8** ✓

**b.** 0.8

**c.** -0.6

**d.** -0.4

In the second quadrant, cosine is negative: $\cos x = -\sqrt{1 - 0.6^2} = -\sqrt{0.64} = -0.8$. [M1 A1]

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